

See What You Hear – Real-time DSP and Embedded LED

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מדעי המחשב
המכללה האקדמית ספיר (נצ)



SEE WHAT YOU HEAR

Project Goal

Enhance existing LED setups by developing a system that dynamically translates live audio features into real-time LED responses.

Architecture

LED Communication

Transmits processed data to LED displays.



GUI

Provides a user interface for interaction and control.



Color Decision

Applies machine learning to interpret frequency data.



FFT Module

Analyzes sound frequencies using Fast Fourier Transform.



Audio Processor

Captures live audio input for processing.



Usage stages & UI

- Open a host Daw
- Select the desired Audio channel to visualize
- Choose the DSP as part of the inserts process chain
- Instance created



Response Curve



Overall Brightness

LED Simulator

Modes

FFT processing chain in audio plugin

Audio Processing Thread

Audio samples (L/R) are pushed into fixed-size blocks



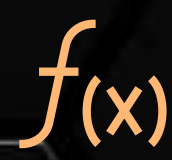
FFT Processor Thread

Pulls pre-processed block and shifts remaining blocks to queue head



FFT Calculation

Performs FFT calculation on windowed data in FFT order



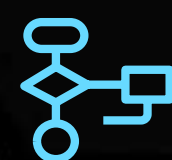
Bin Processing

FFT consumers process frequency bins to extract audio features for analysis.

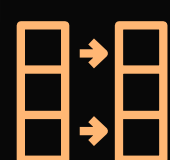


Color Decision

Provides FFT data to Color Decision for processing



FIFO



FFT Windowing



FIFO



GUI Visualization

Fixed-size audio blocks are queued in the FIFO for pre-processing

Applies a windowing function to the data.

Pushes processed blocks into FIFO

Updates frequency response curve in UI

Comparison of LED Systems



Traditional LED



See What You Hear

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✓

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✓

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✓

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✓

Real-time response to live audio

Direct audio integration from DAW

Audio feature-based control

Machine learning-based color mapping

